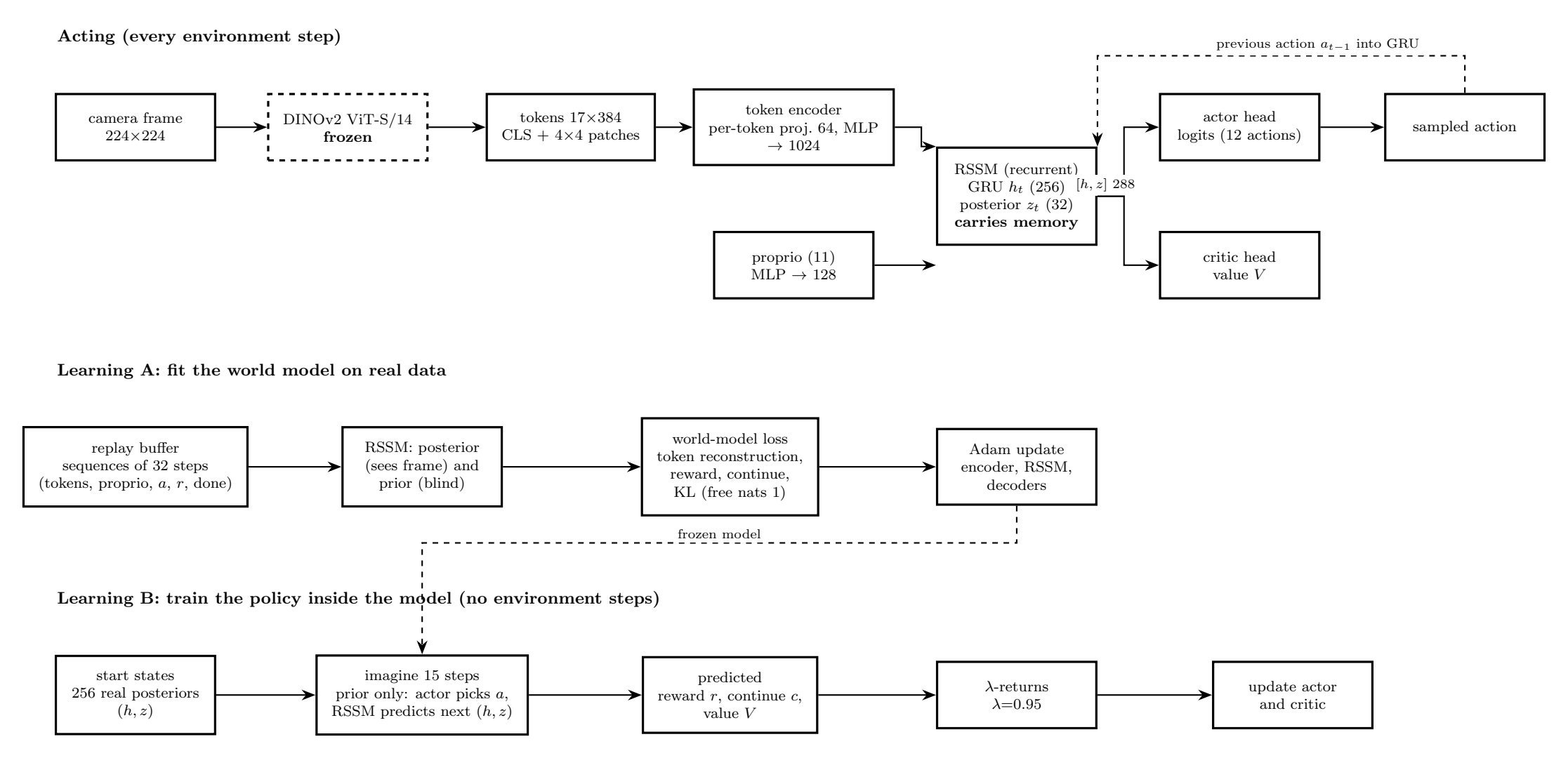

Acting (every environment step)
camera frame 224×224
DINOv2 ViT-S/14 frozen
tokens 17×384 CLS + 4×4 patches
token encoder per-token proj. 64, MLP → 1024
proprio (11) MLP → 128
RSSM (recurrent) GRU h_t (256) posterior z_t (32) carries memory
$[h, z]$ 288
actor head logits (12 actions)
sampled action
critic head value V
previous action a_{t-1} into GRU
Learning A: fit the world model on real data
replay buffer sequences of 32 steps (tokens, proprio, a, r, done)
RSSM: posterior (sees frame) and prior (blind)
world-model loss token reconstruction, reward, continue, KL (free nats 1)
Adam update encoder, RSSM, decoders
frozen model
Learning B: train the policy inside the model (no environment steps)
start states 256 real posteriors (h, z)
imagine 15 steps prior only: actor picks a, RSSM predicts next (h, z)
predicted reward r, continue c, value V
λ-returns λ=0.95
update actor and critic